

SIDO KANHU MURMU UNIVERSITY, DUMKA



FOUR YEAR UG PROGRAMME BOTANY HONOURS/RESEARCH

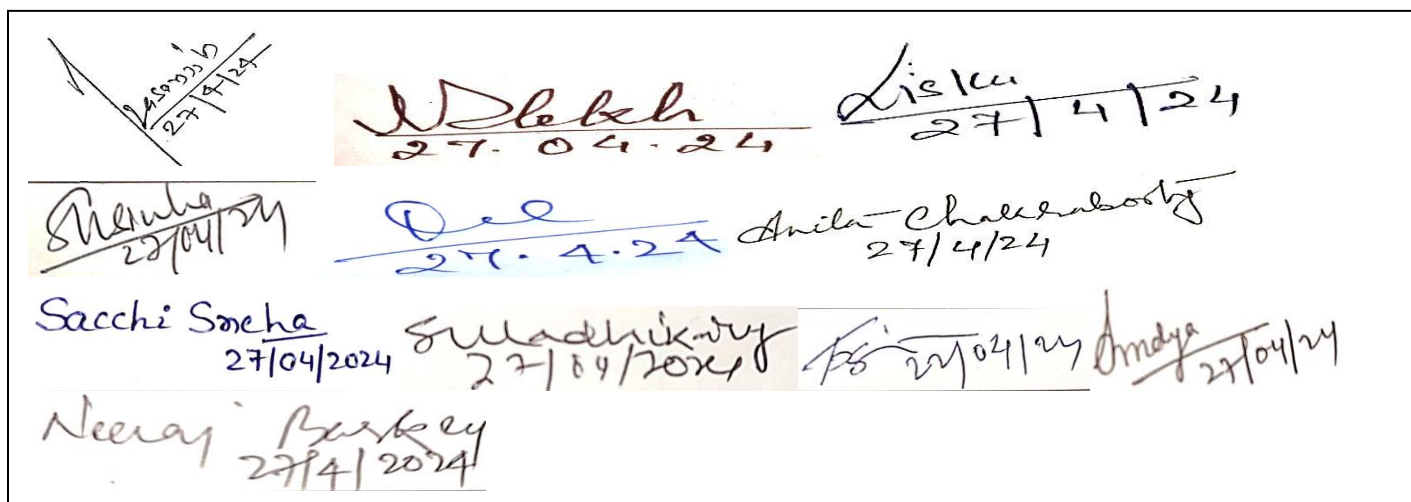
SYLLABUS

UNDER

SIDO KANHU MURMU UNIVERSITY, DUMKA

Implemented from

Academic Session 2023-2027

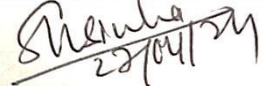
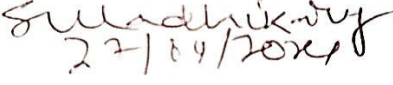
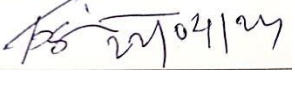
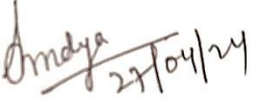
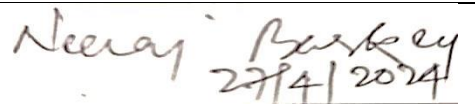
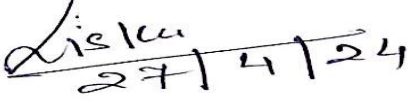
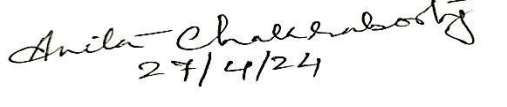
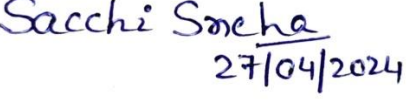


B.SC. (Botany) Syllabus as Per National Education Policy

Meeting Proceedings

An offline/online meeting (Google meet-meet.google.com/yyi-tkvt-ywd) of Board of Courses and Studies to review/modify the syllabi of undergraduate (FYGUP) was held on 27.04.2024 under the chairmanship of Dr. Sanjay Kumar Sinha, Head, University Department of Botany, SKMU, Dumka. The following members (Board of Courses and studies) were present offline/online in this meeting. It was unanimously accepted and recommended the draft copy of syllabi with minor modification.

Members of Board of Studies

SN	Name	Board of Studies	Signature (Physical/Digital)
1	Dr. Sanjay Kumar Sinha Head, University Department of Botany SKMU, Dumka	Chairman	 27/04/24
2	Dr. S.N. Adhikari Dean, Faculty of Science SKM University, Dumka	Member (Invited)	 27/04/2024
3	Dr. T.P. Singh State Nodal Officer (NEP) SKM University, Dumka	Member (Invited)	 27/04/24
4	Dr. P. Mukherjee Assistant Professor & Head Department of Botany, K.O. College, Gumla, Ranchi University, Ranchi	External Member	 27/4/24
5	Dr. Sutanu Lal Bondya Assistant Professor University Department of Botany SKMU, Dumka	Member	 27/04/24
6	Neeraj Baskey Assistant Professor University Department of Botany SKM University, Dumka	Member	 27/4/2024
7	Dr. Monika Patralekh Assistant Professor & Head Department of Botany Deoghar College, Deoghar SKMU, Dumka	Member	 27.04.24
8	Dr. Meera Choudhary Assistant Professor & Head Sahibganj Mahila College, Sahibganj SKMU, Dumka	Member	 27.4.24
9	Dr. Samuel Kisku Assistant Professor & Head Department of Botany S.P. College, Dumka SKM University, Dumka	Member	 27/4/24
10	Dr. Anita Chakraborty Assistant Professor Department of Botany S.P. College, Dumka SKMU, Dumka	Member	 27/4/24
11	Dr. Sacchi Senha Assistant Professor & Head Department of Botany Godda College, Godda SKM University, Dumka	Member	 27/04/2024

Semester	Paper code	Title of the course	Internal	End Semester	Total Marks	Credit	Total Credit
SEM-I	MJ-BOT-01	Microbiology, Algae & Fungi	25	75	100	4	4
	MN-BOT-1	Diversity of Non- Flowering Plants	15	60	75	3	4
		Practical	-	25	25	1	
	MDC-BOT-01/02/03	Introductory Botany	-	75	75	3	3
SEM-II	MJ-BOT-02	Bryophytes, Pteridophytes Paleobotany & Gymnosperms	25	75	100	4	4
	MJ-BOT-03	Practical -I (Practical based on MJ-Bot-02)	25	75	100	4	4
	MDC-BOT-01/02/03	Introductory Botany	-	75	75	3	3
SEM-III	MJ-BOT-04	Morphology of Angiosperms	25	75	100	4	4
	MJ-BOT-05	Practical -II (Practical based on MJ-Bot-04)	25	75	100	4	4
	MN-BOT-2	Morphology & Systematics of Angiosperms	15	60	75	3	4
		Practical	-	25	25	1	
	MDC-BOT-01/02/03	Introductory Botany	-	75	75	3	3
SEM-IV	MJ-BOT-06	Embryology & Anatomy	25	75	100	4	4
	MJ-BOT-07	Economic Botany	25	75	100	4	4
	MJ-BOT-08	Practical -III (Practical based on MJ-Bot-06 and 07)	25	75	100	4	4
SEM-V	MJ-BOT-09	Cell Biology	25	75	100	4	4
	MJ-BOT-10	Molecular Biology	25	75	100	4	4
	MJ-BOT-11	Practical –IV (Practical based on MJ-Bot-09 and 10)	25	75	100	4	4
	MN-BOT-3	Plant Anatomy & Embryology	15	60	75	3	4
		Practical	-	25	25	1	
SEM-VI	MJ-BOT-12	Genetics & Plant Breeding	25	75	100	4	4
	MJ-BOT-13	Plant Physiology and Metabolism	25	75	100	4	4
	MJ-BOT-14	Biochemistry and Bioinformatics	25	75	100	4	4
	MJ-BOT-15	Practical-V (Practical based on MJ-Bot-12, 13 and 14)	25	75	100	4	4
SEM-VII	MJ-BOT-16	Ecology & Environmental Biology	25	75	100	4	4
	MJ-BOT-17	Plant Biotechnology and Tissue Culture	25	75	100	4	4
	MJ-BOT-18	Ethnobotany & Biodiversity	25	75	100	4	4
	MJ-BOT-19	Practical -VI (Practical based on MJ-Bot-16, 17 and 18)	25	75	100	4	4
	MN-BOT-4	Plant Physiology & Metabolism	15	60	75	3	4
		Practical	-	25	25	1	
SEM-VIII	MJ-BOT-20	Dissertation, Seminar / Group Discussion	25	75	100	4	4
	AMJ-BOT-01	Advance Microbiology	25	75	100	4	4
	AMJ-BOT-02	Plant Pathology	25	75	100	4	4
	AMJ-BOT-03	Practical-VII (Practical based on AMJ-BOT-01 & 02)	25	75	100	4	4
SEM-VIII	BOT-RC-01	Research Methodology	Distribution of Marks mentioned in the syllabus		100	4	12
	BOT-RC-02	Research project or Dissertation			200	8	

SEMESTER-I

MJ- BOT- 01 (MICROBIOLOGY, ALGAE & FUNGI)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. This course will provide knowledge on various fields of basic Botany.
2. The syllabus is prepared to enable students for competitive exams in frontier areas of plant sciences.
3. Students will be able to know about habit, habitat, morphology, anatomy and reproduction of various plant groups.

Course Outcome:

1. Students will understand about the classification and diversity of different microbes including Bacteria, Viruses, Algae, Fungi & their economic importance.
2. Students will understand the conceptual skill about identifying microbes & pathogens.

UNIT 1: MICROBIOLOGY

1. **Methods of Microbiology:** Sterilization, Purification & isolation of viruses, Culture and Culture media. **(10 Lectures)**
2. **Viruses:** General account of TMV and Bacteriophage. **(3 Lectures)**
3. **Mycoplasma:** General accounts **(2 Lectures)**
4. **Bacteria:** General characters, structure, reproduction and economic importance **(5 Lectures)**
5. **Cyanobacteria:** General account, reproduction and life history of *Nostoc* & *Oscillatoria* **(5 Lectures)**

UNIT 2: ALGAE AND FUNGI

1. **Algae:** General characters, classification (proposed by Fritsch) and economic importance; important features and life history of *Volvox*, *Oedogonium*, *Chara*, *Sargassum* and *Polysiphonia*. **(20 Lectures)**
2. **Fungi:** General characters, classification (proposed by Alexopoulos) and economic importance; important features and life history of *Phytophthora*, *Peziza*, *Puccinia* and *Agricus*. General account of Lichens. **(15 Lectures)**

Suggested Readings:

1. Dubey, R.C. & Maheswari, D.K. A Text Book of Microbiology, 2005, S.Chand & Company
2. Biswas, S.B. & Biswas, A. An Introduction to Viruses (4th ed.), 1996, Vikas Publishing House
3. Power, C.B. & Dagainawala, H.F. General Microbiology, Vol. I&II, Himalaya Publishing House
4. Kumar, H.D. Introductory Phycology (2nd ed.), 1999, Affiliated East-West Press Pvt. Ltd.
5. Vashishta, B.R., Sinha, A.K. & Singh, V.P. Algae (9th ed.), 2002, S. Chand & Company
6. Dubey, H.C. An Introduction to Fungi (2nd ed.), 1990, Vikas Publishing House 23
7. Sharma, P.D. Fungi & Allied Organisms, 2005, Narosa Publishing House
8. Sharma, O.P. Text book of Fungi, Tata McGraw Hill

MN- BOT- 01 (DIVERSITY OF NON- FLOWERING PLANTS)

- **Total Credits: -03**
- **Total teaching Hours: 45**
- **Total Marks: 75**
- **Semester Internal Examination:15 Marks**
- **End Semester University Examination: 60 Marks**

Course objectives:

1. This course will provide knowledge on various fields of basic Botany.
2. The syllabus is prepared to enable students for competitive exams in frontier areas of plant sciences.
3. Students will be able to know about habit, habitat, morphology, anatomy and reproduction of various plant groups.

Course Outcome:

1. Students will get understand about the classification and diversity of different microbes including Bacteria, Viruses, Algae, Fungi & their economic importance.
2. Students will get understand conceptual skill about identifying microbes & pathogens.
3. Student will Gain knowledge about developing critical understanding on morphology, general characters, life cycles and economic importance of Bryophytes, Pteridophytes and Gymnosperms.

Unit 1: Viruses: General account of TMV and Bacteriophage. **(5 Lectures)**

Unit 2: Bacteria: General characters and economic importance. **(5 Lectures)**

Unit 3: Algae: General characters, economic importance and life history of *Nostoc*, *Oedogonium*, and *Polysiphonia*. **(7 Lectures)**

Unit 4: Fungi: General characters, economic importance and life history of *Phytophthora*, *Peziza* & *Puccinia*. **(7 Lectures)**

Unit 5: Bryophytes: General account, economic importance and life history of *Marchantia*, & *Polytrichum*. **(7 Lectures)**

Unit 6: Pteridophytes: General account and life history of *Selaginella*, *Equisetum* & *Marsilia*. **(7 Lectures)**

Unit 7: Gymnosperm: General account and life history of *Cycas* and *Pinus*. **(7 Lectures)**

Suggested Readings:

1. Roy, S.C. and De, K.K. Cell Biology, 1997, New Central Book Agency.
2. Ganguli, H.C. Das, K.S.K. & Dutta, C.T. College Botany, Vol. I, latest Ed., New Central Book Agency
3. Ganguli, H.C. and Kar, A.K. College Botany, Vol. II, latest Ed., New Central Book Agency
4. Sen, S. 1992. Economic Botany, New Central Book Agency, Kolkata.
5. Verma, V. 1974. A Textbook of Economic Botany, Emcay Publication, New Delhi.
6. Bendre, A., Kumar, A. Economic Botany, Rastogi Publication, New Delhi. India
7. Pandey, B.P. Plant Anatomy, Latest Ed., S. Chand & Company.

PRACTICAL (BASED ON MN-BOT-01)

- **Total Credits: -01**
- **Total teaching Hours: 30**
- **Total Marks: 25 (25 Marks End Semester Practical Examination)**

Unit 1: Microbiology: Staining and identification of bacteria.

Unit 2: Algae: Preparation of temporary slides/permanent slides, enumeration of salient features and identification of *Nostoc*, *Oedogonium* and *Polysiphonia*.

Unit 3: Fungi: Preparation of temporary slides/permanent slides, enumeration of salient features and identification of *Phytophthora*, *Peziza* and *Puccinia*.

Unit 4: Bryophytes: Preparation of temporary slides/permanent slides, enumeration of salient features and identification of *Marchantia*, & *Polytrichum*.

Unit 5: Gymnosperm: Preparation of temporary slides/permanent slides, enumeration of salient features and identification of *Cycas* & *Pinus*.

Distribution of marks (Practical):

Experiments	:	10 marks
Spotting	:	05 marks
Class records/Charts/Model/Herbarium etc.	:	05 marks
Viva voce	:	05 marks

MDC- BOT- 01/02/03 (INTRODUCTORY BOTANY)

- **Total Credits: -03**
- **Total teaching Hours: 45**
- **Total Marks: 75**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. Promote understanding of basic principles of Botany.
2. Encourage learning of emerging knowledge and its relevance to individual and society.
3. Promote rational/scientific attitude towards issues related to population, environment and development.
4. Appreciate that the most complex biological phenomena are built on essentially simple processes.

Course Outcome:

1. The students would get an exposure to various branches of Botany in the curriculum in a more contextual and systematic manner as they study its various units.

Unit 1: Morphology of Angiospermic Plants: General account and modification of flowering plants; Root, Stem, Leaf, Flower, Fruit and Seed. **(7 Lectures)**

Unit 2: Cell the unit of life: Prokaryotic & Eukaryotic cell; Structure and function of cell organelles; Plastids, Mitochondria, Ribosome & Nucleus. **(7 Lectures)**

Unit 3: Anatomy of Angiospermic Plants: Classification & function of simple and complex tissue. General account of dicots and monocots (root & stem). **(7 Lectures)**

Unit 4: Ecology and Environment: Ecosystem; Structure, function & components; Energy flow, Fresh water & Forest ecosystem. Pollution; Air & water pollution and its control. Greenhouse effect, Ozone layer depletion. **(8 Lectures)**

Unit 5: Biodiversity: Concept & levels, biodiversity conservation strategies; in-situ- National parks & Biosphere reserve; Ex-situ- Botanical gardens & Seed bank. **(8 Lectures)**

Unit 6: Economic Botany: General account & importance; Cereals (Rice, wheat & Maize) Pulses (Gram, Pea & Lentil) Timber (Sal, Teak & Sissoo) Medicinal Plants (Neem, Kalmegh & Ashwagandha). **(8 Lectures)**

Suggested Readings:

1. Ganguli, H.C. Das, K.S.K. & Dutta, C.T. College Botany, Vol. I, latest Ed., New Central Book Agency
2. Ganguli, H.C. and Kar, A.K. College Botany, Vol. II, latest Ed., New Central Book Agency
3. Sen, S. 1992. Economic Botany, New Central Book Agency, Kolkata.
4. Verma, V. 1974. A Textbook of Economic Botany, Emcay Publication, New Delhi.
5. Bendre, A., Kumar, A. Economic Botany, Rastogi Publication, New Delhi. India
6. Pandey, B.P. Plant Anatomy, Latest Ed., S. Chand & Company.

SEMESTER-II

MJ- BOT- 02 (BRYOPHYTES, PTERIDOPHYTES, PALEOBOTANY & GYMNOSPERMS)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course Objectives

1. To provide knowledge on various fields of basic Botany.
2. To enable students for competitive exams in frontier areas of plant sciences.
3. To provide knowledge about habit, habitat, morphology, anatomy and reproduction of various plant groups.

Course Outcome: After the completion of the course the students will be able to:

1. Students will understand on morphology, anatomy, classification, economic importance, life history and reproduction of Bryophytes, Pteridophytes & Gymnosperms
2. Students will understand of plant evolution and their transition to land habitat.
3. Students will understand on morphology, anatomy, reproduction in plant fossils, challenges posed for their extinction & plant evolution and their transition to land habitat.

UNIT 1: BRYOPHYTES & PTERIDOPHYTES

1. **Bryophytes:** General account, classification (Cavers and Parihar); progressive sterilization of sporogeneous tissue, economic importance of Bryophytes; Life history of *Marchantia*, *Anthoceros*, *Sphagnum* and *Polytrichum*. **(15 Lectures)**
2. **Pteridophytes:** General account, Classification (Parihar & Smith), anatomy, stellar evolution, Apospory & Apogamy of Pteridophytes Life history of *Psilotum*, *Selaginella*, *Equisetum* & *Marsilia*. **(15 Lectures)**

UNIT 2: PALEOBOTANY & GYMNOSPERMS)

1. **Paleobotany:** Geological time scale, Types, major fossils sites in Jharkhand. General account and life history of *Rhynia*, *Pentoxylum* & *Williamsonia*. **(15 Lectures)**
2. **Gymnosperms:** General account & Classification (Pant & Arnold). Morphology, anatomy, reproduction and economic importance of *Cycas*, *Pinus* & *Gnetum*. **(15 Lectures)**

Suggested Readings:

1. Parihar, N.S. Introduction to Embryophyte (Vol. 1 Bryophyta), Central Book Distributors
2. Vashista, B.R. Bryophyta, Latest Ed., S. Chand & Company
3. Vashista, P.C. Pteridophytes, Latest Ed., S. Chand & Company Pvt. Ltd.
4. Agashe, S.N. Palaeobotany, Latest Ed., Oxford & IBH
5. Dutta, S.C. An Introduction to Gymnosperms (3rd ed.), 1984, Kalyani Publishers

MJ- BOT- 03 (PRACTICAL BASED ON MJ-BOT-02)

- **Total Credits: -04**
- **Total teaching Hours: 120**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Unit 1: Bryophytes:

1. Preparation of temporary slides/permanent slides, enumeration of salient features and identification of *Marchantia*, *Anthoceros*, *Sphagnum* and *Polytrichum*.

Unit 2: Pteridophytes:

1. Preparation of temporary slides/permanent slides, enumeration of salient features and identification of *Psilotum*, *Selaginella*, *Equisetum*, and *Marsilea*.

Unit 3: Paleobotany:

1. Study of temporary slides/permanent slides, enumeration of salient features and identification of *Rhynia*, *Pentoxylum* & *Williamsonia*.

Unit 4: Gymnosperms:

1. Preparation of temporary slides/permanent slides, enumeration of salient features and identification of *Cycas*, *Pinus* & *Gnetum*.

Distribution of marks (Practical):

Experiments	:	30 marks
Spotting	:	10 marks
Class records/Charts/Model/Herbarium etc.	:	20 marks
Viva voce	:	15 marks

SEMESTER-III

MJ- BOT- 04 (MORPHOLOGY AND SYSTEMATICS OF ANGIOSPERMS)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. To learn the morphology and structure and modification of Angiospermic Plants.
2. To learn various field of plant systematics.

Course Outcome:

1. The unit will enable the students to learn about the morphology and structure and modification of Angiospermic Plants.
2. The unit will enable the students to learn about plant systematics.

Unit-1: Morphology of Angiosperms

1. **The Root:** Definition, type, modification and function **(5 Lectures)**
2. **The Stem:** Definition, type, modification and function **(5 Lectures)**
3. **The Leaf:** Definition, parts, venation, types, phyllotaxy, modification and function. **(5 Lectures)**
4. **The Inflorescence:** Definition, different types **(5 Lectures)**
5. **The Flower:** Definition, position, parts and aestivation **(5 Lectures)**
6. **The Fruit:** Definition parts and types **(5 Lectures)**
7. **The Seed:** Definition, Structure of dicotyledonous and Monocotyledonous seed **(5 Lectures)**

Unit 2: Systematic of Angiosperms

1. **Systematic in practice:** Herbarium – Importance, preparation & role; important herbaria & Botanical Gardens of India. **(5 Lectures)**
2. **Botanical nomenclature:** Binomial nomenclature; principles and rules; typification; principle of priority and its limitations. **(5 Lectures)**
3. **System of classification:** Bentham and Hooker's system and Hutchinson system. **(5 Lectures)**
4. **Modern trends in Plant taxonomy:** Taxonomy in relation to Cytology (Cytotaxonomy), Chemotaxonomy and Numerical taxonomy. **(5 Lectures)**
5. **Study of diagnostic features of the families:** *Ranunculaceae*, *Apocynaceae*, *Verbenaceae*, *Lamiaceae*, *Euphorbiaceae*, *Poaceae* and *Cyperaceae*. **(5 Lectures)**

Suggested Readings:

1. Ganguli, H.C., Das, K.S.K. & Dutta, C.T. College Botany, Vol. I, latest Ed., New Central Book Agency
2. Ganguli, H.C. and Kar, A.K. College Botany, Vol. II, latest Ed., New Central Book Agency
3. Mukherjee, S. College Botany, Vol. III, latest Ed., New Central Book Agency
4. Ananta Rao T. Morphology of Angiosperms.

MJ- BOT- 05 (Practical based on MJ-Bot-04)

- **Total Credits: -04**
- **Total teaching Hours: 120**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Unit 1: Morphology and Systematics of Angiosperms

1. Study of Museum specimens of different morphological parts of the plant especially leaves inflorescence, flowers and fruits. Preparation of Herbarium.
2. **Study of diagnostic features of the families:** Study of vegetative and floral characters of the following families (Description, V.S. flower, section of ovary, floral diagram/s, floral formula/e and systematic position according to Bentham & Hooker's system of classification): *Ranunculaceae*, *Apocynaceae*, *Verbenaceae*, *Lamiaceae*, *Euphorbiaceae*, *Poaceae* and *Cyperaceae*.

Distribution of marks (Practical):

Experiments	:	30 marks
Spotting	:	10 marks
Class records/Charts/Model/Herbarium etc.	:	20 marks
Viva voce	:	15 marks

MN- BOT- 02 (MORPHOLOGY & SYSTEMATIC OF ANGIOSPERMS)

- **Total Credits: -03**
- **Total teaching Hours: 45**
- **Total Marks: 75**
- **Semester Internal Examination:15 Marks**
- **End Semester University Examination: 60 Marks**

Course objectives:

1. To learn the morphology and structure and modification of Angiospermic Plants.
2. To learn various field of plant systematics.

Course Outcome:

1. The unit will enable the students to learn about the morphology and structure and modification of Angiospermic Plants.
2. The unit will enable the students to learn about plant systematics.

Unit-1: Morphology of Angiosperms

1. **The Root:** Definition, type, modification and function **(5 Lectures)**
2. **The Stem:** Definition, type, modification and function **(5 Lectures)**
3. **The Leaf:** Definition, parts, venation, types, phylotaxy, modification and function. **(5 Lectures)**
4. **The Inflorescence:** Definition, different types **(5 Lectures)**
5. **The Flower:** Definition, position, parts and aestivation **(6 Lectures)**

Unit 2: Systematic of Angiosperms

1. **Botanical nomenclature:** Binomial nomenclature; principles and rules; typification; principle of priority and its limitations. **(7 Lectures)**
2. **System of classification:** Bentham and Hooker's system and Hutchinson system. **(5 Lectures)**
3. **Study of diagnostic features of the families:** *Apocynaceae, Verbenaceae, Lamiaceae, and Poaceae.* **(7 Lectures)**

Suggested Readings:

1. Ganguli,H.C., Das, K.S.K. & Dutta, C.T. College Botany, Vol. I, latest Ed., New Central Book Agency
2. Ganguli,H.C. and Kar, A.K. College Botany, Vol. II, latest Ed., New Central Book Agency
3. Mukherjee, S. College Botany, Vol. III, latest Ed., New Central Book Agency
4. Ananta Rao T. Morphology of Angiosperms.

PRACTICAL (BASED ON MN-BOT-02)

- **Total Credits: -01**
- **Total teaching Hours: 30**
- **Total Marks: 25 (25 Marks End Semester Practical Examination)**

Unit 1: Morphology and Systematics of Angiosperms

1. **Study of diagnostic features of the families:** Study of vegetative and floral characters of the following families (Description, V.S. flower, section of ovary, floral diagram/s, floral formula/e and systematic position according to Bentham & Hooker's system of classification): *Apocynaceae*, *Verbenaceae*, *Lamiaceae*, & *Poaceae*
2. **Morphological study of :** Root, Stem, Leaf, Inflorescence and Flower

Distribution of marks (Practical):

Experiments	:	10 marks
Spotting	:	05 marks
Class records/Charts/Model/Herbarium etc.	:	05 marks
Viva voce	:	05 marks

SEMESTER-IV

MJ- BOT- 06 (EMBRYOLOGY AND ANATOMY)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. The objective of this paper is to provide basic knowledge of plant anatomy, cellular composition associated with the reproduction and reproduction in plants.
2. This helps them to understand how different plant tissue structure evolve and modify their functions with respect to their environment. Also, to make them aware about embryology.

Course Outcome:

1. After completion of course students shall have adequate knowledge regarding embryology makes they understand how reproduction play significant role in defining population structure and natural diversity.
2. Also, after successful completion of the course the student shall have adequate knowledge about the basic principle of anatomy.

Unit 1: EMBRYOLOGY

1. **Anther and pollen biology:** Microsporogenesis, Microgametogenesis. **(7 Lectures)**
2. **Ovule:** Structure Types; Megasporogenesis and megagametogenesis. **(7 Lectures)**
3. **Pollination and fertilization:** Types and contrivances of self & cross pollination; Process of fertilization & double fertilization. **(7 Lectures)**
4. **Embryo, Endosperm and Seed:** Structure and types; General pattern of development of dicot and monocot embryo and endosperm; Seed types & structure, importance and dispersal mechanisms **(14 Lectures)**
5. **Polyembryony and apomixes:** Introduction; Types & structure; General account of apomixes **(8 Lectures).**

Unit 2: ANATOMY

6. **Tissue:** Meristematic, simple and complex. **(4 Lectures)**
7. **Cambium:** Origin, structure and function. **(4 Lectures)**
8. **Periderm:** Origin, structure and function. **(4 Lectures)**
9. **Secondary growth:** Normal secondary growth in dicot stem; Anomalous secondary growth in *Achyranthus*, *Boerhaavia* and *Dracaena*. **(5 Lectures)**

Suggested Readings:

1. Pandey, B.P. Plant Anatomy, Latest Ed., S. Chand & Company.
2. Tayal, M.S. Plant Anatomy, Latest Ed., Rastogi Publications.
3. Roy, P. Plant Anatomy, Latest Ed., New Central Book Agency.

MJ- BOT- 07 (ECONOMIC BOTANY)

- Total Credits: -04
- Total teaching Hours: 60
- Total Marks: 100
- Semester Internal Examination: 25 Marks
- End Semester University Examination: 75 Marks

Course objectives:

1. To make the students familiar with economic importance of diverse plants that offer resources to human life
2. It emphasizes the plants used as- food for man, fodder for cattle, feed for poultry, plants having medicinal value and also plant source of huge economic value etc.

Course Outcome:

1. After studying Economic Botany, students would have first-hand information of plants used as food, the various kinds of nutrients available in the plants.
2. The students will learn about the use of fiber plants, beverages, fruits and vegetables that are integral to day to day life of plants.
3. Students will learn to explore the regional diversity in food crops and other plants and their ethno-botanical importance as well.

Economic Botany

1. **Food plants:** A general account of Cereals (Wheat, Rice & Maize), Pulses (Pigeon pea, Gram, & Lentil) **(12 Lectures)**
2. **Oil yielding sources:** Edible (Mustard, Sunflower, & Sesame) and Non edible (Castor, Karanj, & Jatropha). **(12 Lectures)**
3. **Plant fibers:** A general account of common fiber plants (Cotton, Jute, Bamboos, & Flax). **(12 Lectures)**
4. **Timber:** A general account (Sal, Teak, Sissoo, Siris & Gambhar). **(10 Lectures)**
5. **Medicinal plants:** A brief account of common medicinal Plants of Jharkhand (Ashwagandha, Sarpagandha, Kalmegh, Centella, Arjuna, Adhatoda, Tulsi, Nux-vomica, Terminalia chebula, T. bellerica & Awanla). **(14 lectures)**

Reference Books:

1. Bhojwani, S.S. and Bhatnagar, S.P. (2011). The Embryology of Angiosperms, Vikas Publishing House. Delhi. 5th edition.
2. Shivanna, K.R. (2003). Pollen Biology and Biotechnology. Oxford and IBH Publishing Co. Pvt. Ltd. Delhi. Raghavan, V. (2000). Developmental Biology of Flowering plants, Springer, Netherlands. 4. Johri, B.M. (1984). Embryology of Angiosperms, Springer-Verlag, Netherlands.
3. Kochhar, S.L. (2012). Economic Botany in Tropics, MacMillan & Co. New Delhi, India.
4. Wickens, G.E. (2001). Economic Botany: Principles & Practices. Kluwer Academic Publishers, The Netherlands.

MJ- BOT- 08 (PRACTICAL BASED ON MJ-BOT-06 AND 07)

- **Total Credits: -04**
- **Total teaching Hours: 120**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Unit 1: Embryology

1. Study of types through permanent slides/ photographs- Structure of anther.
2. Types of ovules
3. Female gametophyte: Polygonum (monosporic) type of Embryo sac Development
4. Pollination types and seed dispersal mechanisms (including appendages, aril, caruncle)
5. Dissection of embryo/endosperm from developing seeds;
6. Calculation of percentage of germinated pollen in a given medium.

Unit 2: Anatomy

1. Study of anatomical details through permanent slides/temporary stain mounts/ macerations/ museum specimens with the help of suitable examples.
 - a. Types of parenchyma, collenchyma and sclerenchyma.
 - b. Xylem: Tracheary elements-tracheids, vessel elements; thickenings; perforation plates; xylem fibres.
 - c. Phloem: Sieve tubes-sieve plates; companion cells; phloem fibers.
 - d. Epidermal system: stomata types.
 - e. Root: monocot, dicot.
 - f. Stem: monocot, dicot - primary and secondary growth.
 - g. Leaf anatomy: isobilateral, dorsiventral.
 - h. Adaptive Anatomy: xerophytes, hydrophytes.

Unit 3: Economic Botany

1. Botanical characters and uses of
 - a. Food plants- Cereals and Pulses
 - b. Oil yielding Plants
 - c. Fibers yielding
 - d. Major timber yielding
 - e. Medicinal plants

Distribution of marks (Practical):

Experiments	:	30 marks
Spotting	:	10 marks
Class records/Charts/Model/Herbarium etc.	:	20 marks
Viva voce	:	15 marks

SEMESTER-V

MJ- Bot- 09 (CELL BIOLOGY)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. Cell biology study will provide inside into the organization of cell, its features and regulation at different levels.
2. Through the study of cell organelles, they will be able to understand the various metabolic processes such as respiration, photosynthesis etc.
3. The objective of the present course content is to provide a foundation and background in cellular and acellular entities of plants.

Course Outcome:

1. This course will be able to demonstrate foundational knowledge in understanding of cell.
2. Understanding of Cell metabolism, chemical composition, physiochemical and function.
3. Contemporary approaches in modern cell and molecular biology.

CELL BIOLOGY

1. **The cell:** Cell as a unit of structure and function; Characteristics of prokaryotic and eukaryotic cells. **(12 Lecture)**
2. **Cell wall and plasma membrane:** Chemistry, structure and function of plant cell wall and plasma membrane with special reference to fluid mosaic model. **(12 Lectures)**
3. **Cell organelles :** Structural organization and Function of **(14 Lectures)**
 - a. Chloroplast,
 - b. Mitochondria
 - c. Endoplasmic Reticulum
 - d. Golgi Apparatus and
 - e. Nucleus
4. **Cell division:** Cell cycle, mitosis and meiosis. **(12 Lectures)**
5. **Chromosome:** Morphology, structure, nucleosome model, chemical composition, polytene lampbrush and β -chromosome. **(10 Lectures)**

Suggested readings:

1. Karp, G. (2010). Cell Biology, John Wiley & Sons, U.S.A. 6th edition.
2. Hardin, J., Becker, G., Skliensmith, L.J. (2012). Becker's World of the Cell, Pearson Education Inc. U.S.A. 8th edition.
3. Cooper, G.M. and Hausman, R.E. (2009) The Cell: A Molecular Approach. 5th edition. ASM Press & Sunderland, Washington, D.C.; Sinauer Associates, MA.
4. Becker, W.M., Kleinsmith, L.J., Hardin. J. and Bertoni, G. P. (2009) The World of the Cell. 7th edition. Pearson Benjamin Cummings Publishing, San Francisco.

MJ- BOT- 10 (MOLECULAR BIOLOGY)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. To gain the knowledge of structure and functions of DNA and RNA and its functioning at cellular (molecular) level

Course Outcome: This course will provide-

1. Understanding of nucleic acid, organization of DNA in prokaryotes and Eukaryotes, DNA replication mechanism, genetic code and transcription process.
2. Processing and modification of RNA and translation process, function and regulation of expression. Application in biotechnology.

Unit-1: Molecular Biology

1. **The Structures of DNA and RNA:** The Structures and function of DNA and RNA. Watson and Crick model, Salient features of double helix, Organization of DNA. RNA Structure types & functions. The Nucleosome Chromatin structure- Euchromatin, Heterochromatin- Constitutive and Facultative heterochromatin. **(15 Lectures)**
2. **The replication of DNA:** Types and mechanism of Replication of DNA. **(10 Lectures)**
3. **Genetic code:** Genetic code **(4 Lectures)**
4. **Transcription:** Transcription Concept of central dogma, Transcription in prokaryotes and eukaryotes. Gene silencing. **(8 Lectures)**
5. **Processing and modification of RNA** Split genes-concept of introns and exons, removal of introns, spliceosome machinery, splicing. **(7 Lectures)**
6. **Translation:** Ribosome structure and assembly, mRNA; aminoacyl tRNA synthetases; Various steps in protein synthesis, mechanism of elongation and termination of polypeptides. **(16 Lectures)**

Suggested readings:

1. Watson J.D., Baker, T.A., Bell, S.P., Gann, A., Levine, M., Losick, R. (2007). Molecular Biology of the Gene, Pearson Benjamin Cummings, CSHL Press, New York, U.S.A. 6th edition.
2. Klug, W.S., Cummings, M.R., Spencer, C.A. (2009). Concepts of Genetics. Benjamin Cummings. U.S.A. 9th edition.
3. Russell, P. J. (2010). i-Genetics- A Molecular Approach. Benjamin Cummings, U.S.A. 3rd edition.
4. Griffiths, A.J.F., Wessler, S.R., Carroll, S.B., Doebley, J. (2010). Introduction to Genetic Analysis. W. H. Freeman and Co., U.S.A. 10th edition.
5. Gupta, P.K
6. Snustad, D.P. and Simmons, M.J. (2010). Principles of Genetics. John Wiley and Sons, U.K. 5th edition.

MJ- BOT- 11 (PRACTICAL BASED ON MJ-BOT-09 AND 10)

- **Total Credits: -04**
- **Total teaching Hours: 120**
- **Total Marks: 100**
- **Semester Internal Examination:25 Marks**
- **End Semester University Examination: 75 Marks**

Unit 1 Cell Biology:

1. Study of cell and its organelles.
2. Preparation and types of stains used in cytogenetics (Acetocarmine).
3. Preparation of temporary slides to study different stages of mitosis (Onion root tip/Provided material) and meiosis (Onion floral buds/Provided materials) using squash technique.
4. Study of plant cell structure with the help of epidermal peel mount of Onion/Rhoeo/Crinum.
5. Measurement of cell size by the technique of micrometry.

Unit 2 Molecular Biology:

1. Isolation of DNA & RNA

Distribution of marks (Practical):

Experiments	:	30 marks
Spotting	:	10 marks
Class records/Charts/Model/Herbarium etc.	:	20 marks
Viva voce	:	15 marks

MN- BOT- 03 (PLANT ANATOMY & EMBRYOLOGY)

- **Total Credits: -03**
- **Total teaching Hours: 45**
- **Total Marks: 75**
- **Semester Internal Examination:15 Marks**
- **End Semester University Examination: 60 Marks**

Course Objectives:

- 1.The Objective of this paper is to provide basic knowledge of plant internal architecture, cellular composition and reproduction.
2. This helps them to understand how different plant tissue structure evolve and modify their functions with respect to their environment. Also, to make them aware about identification, nomenclature and classification.

Course Learning Outcomes:

1. Knowledge regarding anatomy equipped the students to identify different types of tissues and make them able to correlate their physiology in a better away.
2. This will also help them to understand how different plant tissue evolve and modify their structure and functions with respect to their environment.
3. Knowledge regarding embryology make them understand how reproduction play significant role in defining population structure and natural diversity.

Unit 1: Meristematic and permanent tissues; Simple and complex tissues **(7 Lectures)**

Unit 2: Secondary Growth Vascular cambium – structure and function. Secondary growth in dicot stem, (heartwood and sapwood) **(10 Lectures)**

Unit 3: General account of adaptations in xerophytes and hydrophytes. **(6 Lectures)**

Unit 4: Structure and types of ovules; Megasporogenesis; Types of embryo sacs, organization and ultrastructure of mature embryo sac, Microsporogenesis and male gametophytes **(6 Lectures)**

Unit 5: Pollination and fertilization Pollination mechanisms and adaptations; Double fertilization; Seed structure dicot and monocot. **(6 Lectures)**

Unit 6: Embryo and endosperm: types, structure and functions **(5 Lectures)**

Unit 7: Apomixis and polyembryony Definition, types and Practical applications **(5 Lectures)**

Suggested readings:

1. Bhojwani, S.S. & Bhatnagar, S.P. (2011). Embryology of Angiosperms. Vikas Publication House Pvt. Ltd New Delhi. 5th edition.
2. Mauseth, J.D. (1988). Plant Anatomy. The Benjamin/Cummings Publisher, USA.

PRACTICAL (BASED ON MN-BOT-03)

- **Total Credits: -01**
- **Total teaching Hours: 30**
- **Total Marks: 25 (25 Marks End Semester Practical Examination)**

Practical:

1. Study of meristems through permanent slides and photographs.
2. Stem: Monocot: Zea mays; Dicot: Helianthus; Secondary: Helianthus (only Permanent slides).
3. Root: Monocot: Zea mays; Dicot: Helianthus; Secondary: Helianthus (only Permanent slides).
4. Leaf: Dicot and Monocot leaf (only Permanent slides).
5. Adaptive anatomy: Xerophyte (Nerium leaf); Hydrophyte (Hydrilla stem).
6. Types of ovules: anatropous, orthotropous, circinotropous, amphitropous/ campylotropous.
7. Female gametophyte: Polygonum (monosporic) type of Embryo sac Development (Permanent slides/photographs).
8. Dissection of embryo/endosperm from developing seeds.

Distribution of marks (Practical):

Experiments	:	10 marks
Spotting	:	05 marks
Class records/Charts/Model/Herbarium etc.	:	05 marks
Viva voce	:	05 marks

SEMESTER-VI

MJ- BOT- 12 (GENETICS & PLANT BREEDING)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. The paper will deal with heredity inheritance pattern among the organism.
2. Linkage and genetic recombination, Gene mapping, Chromosomal structure.
3. The paper also deals with Plant Breeding and its role in Crop improvement.

Course Outcome:

1. The unit will enable the students to learn about the use of linkage and recombination frequencies to map genes, plant breeding practices and its role in crop improvement.
2. The unit will provide an understanding of: - Morphology of chromosomes and its relevance in genetics. - Chromosomal and their role in genome evolution with special reference to crop plants.

Unit-1: Genetics

1. **Mendelism:** Mendel's law of inheritance, back cross and test cross. **(6 Lectures)**
2. **Interaction of genes:** Complementary; supplementary; duplicate and epistatic factors. **(6 Lecture)**
3. **Linkage and recombination:** Coupling and repulsion; two and three point test crosses with their significance in chromosome mapping; theories, mechanism and significance of recombination. **(6 Lectures)**
4. **Determination of sex:** Mechanism of sex determination in Plants. **(6 Lectures)**
5. **Maternal influence of inheritance:** Cytoplasmic inheritance in *Mirabilis* & snail. **(6 Lectures)**
6. **Gene mutation:** Molecular basis of gene mutation; frame shift mutation and substitution mutation; spontaneous and induced mutation. **(6 Lectures)**
7. **Structural changes in chromosomes:** Origin, types and effects of duplications, deletions, inversions and translocations. **(6 Lectures)**
8. **Numerical changes in chromosomes:** Origin, types and effects of auto and allopolyploidy, origin and meiosis of aneuploidy. **(6 Lectures)**

Unit-2: Plant Breeding

1. **Plant breeding:** Aim and objectives, vegetative and sexual breeding. **(6 Lectures)**
2. **Methods of plant improvement:** Introduction and acclimatization, selection – pure line and mass selection, hybridization in self and cross pollinated crops, hybrid vigour. **(6 Lectures)**

Suggested readings:

1. P K Gupta, Plant Breeding, Rastogi Publication.
2. B D Singh, Plant Breeding, Kalyani Publication.
3. Kumar and Sinha, A cytogenetics plant breeding and evolutionary biology.

MJ- BOT- 13 (PLANT PHYSIOLOGY AND METABOLISM)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. The course aims at making students realize how plants function, namely the importance of water, minerals, hormones, and light in plant growth and development; understand transport mechanisms and translocation in the phloem, and appreciate the commercial applications of plant physiology.
2. Current understanding of regulation and integration of metabolic processes in plants with reference to crop productivity. To gain the knowledge of physiological and biochemical processes in the plant system.

Course Outcome:

1. To understand water and nutrient uptake and movement in plants, role of mineral elements, translocation of sugars. Role of various plant growth regulator as, phytochrome cytochromes and phototropins, and flowering stimulus. Students will gain the knowledge on reproductive strategies in higher plants along with physiology of flowering, molecular and hormonal basis of flowering mechanism.

Unit 1 PHYSIOLOGY

1. **Plant – Water relation:** Diffusion and osmosis; water potential and chemical potential; absorption of water; Ascent of sap; transpiration and its significance; mechanism of stomatal movement. **(8 Lectures)**
2. **Mineral nutrition:** Role of macro and micro nutrients in plants. Translocation of solutes, mechanism of mineral salt absorption. **(8 Lectures)**

Unit 2 METABOLISMS

1. **Photosynthesis:** Photochemical reactions; Photophosphorylation; Calvin cycle; C4 cycle and Photorespiration. **(10 Lectures)**
2. **Respiration:** Glycolysis; TCA cycle; Oxidative phosphorylation. **(10 Lectures)**
3. **Nitrogen metabolism:** Biological nitrogen fixation. **(8 Lectures)**
4. **Growth and development:** Physiology of Photoperiodism and vernalization. **(8 Lectures)**
5. **Phytohormones:** Role and mechanism of action of the Phytohormones; Auxin, Cytokinins and Gibberellins. **(8 Lectures)**

Suggested readings:

1. Hopkins, W.G. and Huner, A. (2008). Introduction to Plant Physiology. John Wiley and Sons. U.S.A. 4th edition.
2. Taiz, L., Zeiger, E., Møller, I.M. and Murphy, A (2015). Plant Physiology and Development. Sinauer Associates Inc. USA. 6th edition.
3. Bajracharya D. (1999). Experiments in Plant Physiology-A Laboratory Manual. Narosa Publishing House, New Delhi.
4. Campbell, MK (2012) Biochemistry, 7th ed., Published by Cengage Learning

MJ- BOT- 14 (Biochemistry and Bioinformatics)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. This course will be able to demonstrate foundational knowledge in understanding of cell biochemical level.
2. To familiarize the students with the fundamental principles of Bioinformatics and Computational biology.

Course Outcome:

1. The unit will enable the students to learn about the biochemistry of Cell.
2. Ability to carry out research /investigation independently in specialized area of Bioinformatics and Computational Biology.

Biochemistry

1. **Carbohydrate:** Classification; structure of some representative examples of monosaccharide's disaccharides and polysaccharides. **(8 Lectures)**
2. **Lipid:** Saturated and unsaturated fatty acids; fatty acid biosynthesis; oxidation of fatty acids. **(8 Lectures)**
3. **Nucleic acids:** Composition of nucleic acids; DNA structure and replication. Different forms of RNA and their role. **(6 Lectures)**
4. **Enzymes:** Characteristic, types, classification and mode of action. **(5 Lectures)**
5. **Amino acid and protein metabolism:** Structure, characteristic and classification of amino acids; protein and non-protein amino acids; structure of proteins & protein biosynthesis. **(10 Lectures)**

Bioinformatics

1. **Bioinformatics:** Introduction, genomics, transcriptome, proteome. **(5 Lectures)**
2. **Biological databases:** Generalized and specialized databases – DNA, protein and carbohydrate databases – nucleic acid sequence databases. **(10 Lectures)**
3. Basic concepts of hypothesis testing, two kinds of error, level significance, p value, t- Test for mean and difference between two means, partial t-test., and Chi square test for goodness of fit. **(8 Lectures)**

Suggested readings:

1. Watson J.D., Baker, T.A., Bell, S.P., Gann, A., Levine, M., Losick, R. (2007). Molecular Biology of the Gene, Pearson Benjamin Cummings, CSHL Press, New York, U.S.A. 6th edition. 2.
2. Klug, W.S., Cummings, M.R., Spencer, C.A. (2009). Concepts of Genetics. Benjamin Cummings. U.S.A. 9th edition.

MJ- BOT- 15 (PRACTICAL BASED ON MJ-BOT-12, 13 AND 14)

- **Total Credits: -04**
- **Total teaching Hours: 120**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Unit-1: Genetics

1. Mendel's laws through seed ratios.
2. Laboratory exercises in probability and chi-square.
3. Chromosome mapping using point test cross data.
4. Incomplete dominance and gene interaction through seed ratios. (9:7, 9:6:1, 13:3, 15:1, 12:3:1, 9:3:4).

Unit-2: Plant Breeding

1. Charts as per theory syllabus. (As per avail charts)
 - a. Emasculation, Bagging, Tagging
 - b. Pedigree / Bulk method of breeding,
 - c. Back cross method
2. Demonstration of hybridization techniques (Emasculation, Bagging and tagging)

Unit 3: Physiology and Metabolism

1. Determination of osmotic potential of plant cell sap by plasmolytic method.
2. Determination of water potential of given tissue (potato tuber) by weight method.
3. Study of the effect of wind velocity and light on the rate of transpiration in excised twig/leaf.
4. Calculation of stomatal index and stomatal frequency from the two surfaces of leaves of a mesophyte and xerophyte.
5. Perform rate of photosynthesis and oxygen evolution by Wilmott's bubbler. Perform Moll's experiment.

Unit 4: Biochemistry and Bioinformatics

1. Study the effect of organic solvent and temperature on membrane permeability.
2. Biochemical test of carbohydrate, lipid and protein.
3. Gene identification by using Gene bank (NCBI).
4. Sequence alignment and construction of phylogenetic tree by using tools (BLAST, MEGA, Bio edit).
5. Student t-test and Chi square test.

Distribution of marks (Practical):

Experiments	:	30 marks
Spotting	:	10 marks
Class records/Charts/Model/Herbarium etc.	:	20 marks
Viva voce	:	15 marks

SEMESTER-VII

MJ- BOT- 16 (Ecology and Environmental Biology)

- **Total Credits: 04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. This course aims to introduce the students to the concepts and principles of ecology, biological diversity, conservation, population, community and ecosystem structure and function.
2. To make them understand complex community patterns, processes, and ecosystem functioning.

Course Outcome:

1. It will acquaint the students with complex interrelationship between organisms and environment; and understand methods to studying vegetation, community patterns and processes, ecosystem functions.
2. What are the characteristics of organisms as population, community and ecosystems? What are the ecosystem functions?

Unit-1 Ecology

1. **Ecological adaption:** Hydrophytes and xerophytes. **(7 Lectures)**
2. **Population:** Concept, density and pattern, population growth and population interactions. **(6 Lectures)**
3. **Community:** Community characteristics and their measurement; species diversity; (alpha, beta and gamma). **(8 Lectures)**
4. **Ecosystem:** Structure, function and components of ecosystem; Ecological Pyramids; Energy and its flow in ecosystem; Diversity of ecosystem; Aquatic (fresh water); terrestrial (forest/grassland); Biogeochemical cycles. **(8 Lectures)**

Unit-2 Environmental Biology

1. **Phytogeography:** Introduction; endemism, an account of vegetation of India. **(7 Lectures)**
2. **Ecological management:** renewable and non-renewable natural resources and their management. **(8 Lectures)**
3. **Impact of human activities:** Pollution- Air, Water and Soil pollution; prevention and control of pollution; global warming and ozone depletion. **(8 Lectures)**
4. **Bioremediation:** Important legislations related to environmental management. **(8 Lectures)**

Suggested Readings:

1. Raziuddin, M., Mishra P.K. 2014, A Handbook of Environmental Studies, Akanaksha Publications, Ranchi.
2. Mukherjee, B. 2011: Fundamentals of Environmental Biology. Silverline Publications, Allahabad.
3. Rao, M.N. & Datta, A.K. 1987. Waste Water Treatment. Oxford and IBH Publishing Co. Pvt. Ltd.
4. Raven, P.H., Hassenzahl, D.M. & Berg, L.R. 2012. Environment. 8th edition. John Wiley & Sons.
5. Singh, J.S., Singh, S.P. and Gupta, S.R. 2014. Ecology, Environmental Science and Conservation. S. Chand Publishing, New Delhi.
6. Sharma, P.D. (2010). Ecology and Environment. Rastogi Publications, Meerut, India. 8th edition.

MJ-BOT-17 (PLANT BIOTECHNOLOGY AND TISSUE CULTURE)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course Objectives:

1. The objective of the course is to give students new knowledge and widening of the knowledge acquired in other course by handling of classical and modern plant biotechnology processes, including tissue culture for healthy plants, plants with improved characteristics.
2. This course explores the use of biotechnology to both generate genetic variation in plants and to understand how factors at the cellular level contribute to the expression of genotypes and hence to phenotypic variation.
3. Understanding of biotechnological processes such as recombinant DNA technology.

Learning Outcomes:

1. Learn the basic concepts, principles and processes in plant biotechnology. Have the ability of explanation of concepts, principles and usage of the acquired knowledge in biotechnological and agricultural applications.
2. Use basic biotechnological techniques to explore molecular biology of plants.
3. Understand, how biotechnology is used to for plant improvement and discuss the biosafety concern and ethical issue of that use.

Unit 1: Plant Tissue Culture

1. Introduction, types and composition of media; Nutrient and hormone requirements (role of vitamins and hormones); Totipotency **(5 Lectures)**
2. Organogenesis; Embryogenesis (somatic and zygotic) **(3 Lectures)**
3. Protoplast isolation, culture and fusion **(2 Lectures)**
4. Tissue culture applications (micropropagation, androgenesis, virus elimination, secondary metabolite production, haploids, triploids and hybrids; Cryopreservation; Germplasm Conservation). **(5 Lectures)**

Unit 2: Recombinant DNA technology:

1. Restriction Endonucleases; Restriction Mapping (Linear and Circular); Cloning Vectors: Prokaryotic (pUC 18 and pUC19, pBR322, Ti plasmid, BAC). **(10 lectures)**
2. Recombinant DNA technology, PCR mediated gene cloning; construction of genomic and cDNA libraries, screening DNA libraries to obtain gene of interest by genetic selection; complementation, colony hybridization; PCR. **(10 lectures)**
3. Methods of gene transfer Biological method (Indirect): Agrobacterium-mediated; Physical methods (Direct): Electroporation, Microinjection, Microprojectile bombardment; Selection of transgenics– selectable marker and reporter genes (Luciferase, GUS, GFP). **(10 lectures)**

4. Applications of Biotechnology Pest resistant (Bt-cotton); herbicide resistant plants; Transgenic crops with improved quality traits (Flavr Savr tomato, Golden rice); Improved horticultural varieties; Role of transgenics. **(15 lectures)**

Suggested Readings:

1. Watson J.D., Baker, T.A., Bell, S.P., Gann, A., Levine, M., Losick, R. (2007). Molecular Biology of the Gene, Pearson Benjamin Cummings, CSHL Press, New York, U.S.A. 6th edition.
2. Snustad, D.P. and Simmons, M.J. (2010). Principles of Genetics. John Wiley and Sons Inc., U.S.A. 5th ed.
3. Klug, W.S., Cummings, M.R., Spencer, C.A. (2009). Concepts of Genetics. Benjamin Cummings. U.S.A. 9th edition.
4. Russell, P. J. (2010). i-Genetics- A Molecular Approach. Benjamin Cummings, U.S.A. 3rd edition.
5. Griffiths, A.J.F., Wessler, S.R., Carroll, S.B., Doebley, J. (2010). Introduction to Genetic Analysis.
6. W. H. Freeman and Co., U.S.A. 10th edition.
7. Bhojwani, S.S. and Razdan, M.K., (1996). Plant Tissue Culture: Theory and Practice. Elsevier Science Amsterdam. The Netherlands.
8. Glick, B.R., Pasternak, J.J. (2003). Molecular Biotechnology- Principles and Applications of recombinant DNA. ASM Press, Washington.
9. Bhojwani, S.S. and Bhatnagar, S.P. (2011). The Embryology of Angiosperms. Vikas Publication House Pvt. Ltd., New Delhi. 5th edition.
10. Snustad, D.P. and Simmons, M.J. (2010). Principles of Genetics. John Wiley and Sons, U.K. 5th edition.
11. Stewart, C.N. Jr. (2008). Plant Biotechnology & Genetics: Principles, Techniques and Applications. John Wiley & Sons Inc. U.S.A.

MJ-BOT-18 (ETHNOBOTANY AND BIODIVERSITY)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course Objectives:

1. The objective of the course is to give students new knowledge on the significance of Ethnobotany and application of the herbal technology in India.
2. To learn about the biodiversity richness in global scale and biogeography of India.

Course Outcomes:

1. The students will have holistic understanding on the significance Ethnobotany and application of the herbal technology in India.
2. Get a deep knowledge on biodiversity richness in global scale and biogeography of India.
3. Assess the value of biodiversity wealth of our Nation.
4. Analyze various threats to our biodiversity and able to suggest measures for conservation Strategies.

UNIT-1: ETHNOBOTANY

1. Ethnobotany: Introduction, concept, scope and objectives, relevance of Ethnobotany. **(4 Lectures)**
2. Major and minor ethnic groups or Tribals of Jharkhand. **(4 Lectures)**
3. Methodology of Ethnobotanical studies and plants used by the tribals. **(5 Lectures)**
4. Role of ethnobotany in modern Medicine: Medico-ethnobotanical sources in India. **(5 Lectures)**
5. Role of ethnic groups in conservation of plant genetic resources. Endangered taxa and forest management (participatory forest management). **(6 Lectures)**
6. Ethnobotany and legal aspects Ethnobotany as a tool to protect interests of ethnic groups. Sharing of wealth concept with few examples from India. Biopiracy, Intellectual Property Rights and Traditional Knowledge. **(6 Lectures)**

UNIT-2: BIODIVERSITY

1. **Introduction-** Definition, Genetic diversity, Species diversity, Ecosystem diversity: Structural and functional aspects. **(6 Lectures)**
2. **Value of Biodiversity-** Intrinsic, consumptive, productive use, social, ethical, aesthetic and option values. Biodiversity at global, national and local levels India as a Mega Diversity Nation. Hotspots of Biodiversity. **(6 Lectures)**
3. **Threats to Biodiversity-** Habitat loss, pollution, species introduction, global climate change, overexploitation, poaching of wildlife. **(6 Lectures)**
4. **Conservation of Biodiversity-** Strategies for conservation: In-situ and ex-situ conservation-protected areas-biosphere reserves, sacred groves, National parks and sanctuaries. Ex situ

conservation-Botanical garden, Seed bank, Field gene bank and in vitro conservation.
(6 Lectures)

5. **Conservation Practices in India and World-** Organizations involved in resource conservation IUCN, UNESCO, DBT, NBPGR and BSI. **(6 Lectures)**

Suggested Readings:

1. Krishnamoorthy, K.V (2004) An Advanced text book on Biodiversity- principles and Practice: Oxford and IBH publishing company Pvt. Ltd. New Delhi. 9
2. Kumar, U. and Sharma, A.K (2008) Plant Biotechnology and Biodiversity conservation. Agrobios India.
3. Negi, S.S. (1993) Biodiversity and its Conservation in India. Novacek, M. J. (ed.) (2001) The Biodiversity Crisis: Losing What Counts. New York: American Museum of Natural History Books.
4. Pal, D. K. (2019). Ecosystem Services and Tropical Soils of India. Springer.
5. Sharma P.D. (2003). Ecology and Environmental Sciences, Rastogi Publications, Meerut, India
6. Swaminathan, M.N. & Jain, R.S. (1982) Biodiversity: Implications for global security, Macmillan.
7. Trivedi, P. C. (2007) Global Biodiversity status and conservation. Pointer publishers Jaipur India.
8. S.K. Jain, Manual of Ethnobotany, Scientific Publishers, Jodhpur, 1995.
9. S.K. Jain (ed.) Glimpses of Indian. Ethnobotny, Oxford and I B H, New Delhi – 1981.
10. S.K. Jain (ed.) 1989. Methods and approaches in ethnobotany. Society of ethnobotanists, Lucknow, India.
11. S.K. Jain, 1990. Contributions of Indian ethnobotny. Scientific publishers, Jodhpur. 6)
12. Trivedi P C, 2006. Medicinal Plants: Ethnobotanical Approach, Agrobios, India.
13. Glossary of Indian medicinal plants, R.N.Chopra, S.L.Nayar and I.C.Chopra, 1956. C.S.I.R, New Delhi

MJ- BOT- 19 (PRACTICAL BASED ON MJ-BOT-16, 17 AND 18)

- **Total Credits: -04**
- **Total teaching Hours: 120**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Unit 1: Ecology & Environmental Biology

1. Determination of pH of various soil and water samples (pH meter and pH paper)
2. Comparison of water holding capacity, porosity and rate of infiltration of water in soils of three habitats.
3. Determination of minimal quadrat size for the study of herbaceous vegetation in the college campus, by species area curve method (species to be listed).
4. Quantitative analysis of herbaceous vegetation in the college campus for frequency and comparison with Raunkiaer's frequency distribution law.
5. Quantitative analysis of herbaceous vegetation for density and abundance in the college campus.

Unit 2: Biodiversity

1. Field Survey in Santal Pargana for studying plant species diversity.
2. Ex- situ conservation of plant species using in vitro technique.

Unit 3: Plant Biotechnology & Tissue Culture

1. Preparation of media for tissue culture.
2. Establishment of callus cultures –from carrot.
3. Protoplast isolation and culture.
4. Anther culture.

Unit 4: Ethnobotany and Biodiversity

1. Identification of major Ethnic groups of Jharkhand.
2. Identification of the different categories of Ethno botanical plants.
3. Identification of locally available Medicinal plants.
4. Herbarium preparation of Ethnobotanical, medicinal and aromatic plants.
5. Visit to Botanical/medicinal garden, forest, Wildlife Sanctuary etc.

Distribution of marks (Practical):

Experiments	:	30 marks
Spotting	:	10 marks
Class records/Charts/Model/Herbarium etc.	:	20 marks
Viva voce	:	15 marks

MN- BOT- 04 (PLANT PHYSIOLOGY & METABOLISM)

- **Total Credits: -03**
- **Total teaching Hours: 45**
- **Total Marks: 75**
- **Semester Internal Examination: 15 Marks**
- **End Semester University Examination: 60 Marks**

Course Objectives:

1. The course aims at making students realize how plants function, namely the importance of water, minerals, hormones, and light in plant growth and development.
2. Understand transport mechanisms and translocation in the phloem, and appreciate the commercial applications of plant physiology.

Course Learning Outcomes:

1. The students are able to correlate morphology, anatomy, cell structure and biochemistry with plant functioning
2. The link between theory and practical syllabus is established, and the employability of youth would be enhanced.

Unit 1: Plant-water relations Importance of water, water potential and its components; Transpiration types and its mechanism, significance. **(5 lectures)**

Unit 2: Mineral nutrition Essential elements, macro and micronutrients. **(5 lectures)**

Unit 3: Translocation in phloem & Mechanism of translocation **(5 lectures)**

Unit 4: Photosynthesis- Photosystem I and II, reaction center, antenna molecules; Photophosphorylation; C3 and C4. **(5 lectures)**

Unit 5: Respiration Glycolysis, anaerobic respiration, TCA cycle; Oxidative phosphorylation. **(5 lectures)**

Unit 6: Enzymes Structure and properties; Mechanism and mode of enzyme action. **(5 lectures)**

Unit 7: Nitrogen metabolism Biological nitrogen fixation. **(5 lectures)**

Unit 8: Plant growth regulators Discovery and physiological roles of Auxins, Gibberellins & Cytokinins. **(5 lectures)**

Unit 9: Photoperiodism (SDP, LDP, Day neutral plants); Vernalization. **(5 lectures)**

Suggested Readings:

1. Hopkins, W.G., Huner, N.P., (2009). Introduction to Plant Physiology. John Wiley & Sons, U.S.A. 4th ed.
2. Bajracharya, D., (1999). Experiments in Plant Physiology- A Laboratory Manual. Narosa Publishing House, New Delhi.
3. Bhojwani, S.S. and Razdan, M.K., (1996). Plant Tissue Culture: Theory and Practice. Elsevier Science Amsterdam. The Netherlands.
4. Glick, B.R., Pasternak, J.J. (2003). Molecular Biotechnology- Principles and Applications of recombinant DNA. ASM Press, Washington.

PRACTICAL (BASED ON MN-BOT-04)

- **Total Credits: -01**
- **Total teaching Hours: 30**
- **Total Marks: 25 (25 Marks End Semester Practical Examination)**

1. Determination of osmotic potential of plant cell sap by plasmolytic method.
2. Measurement of rate of transpiration; Farmers photometer/Ganogs photometer.
3. Calculation of stomatal index and stomatal frequency of a mesophyte and a xerophyte.
4. Demonstration of Hill reaction.
5. To study the effect of light intensity and bicarbonate concentration on O₂ evolution in photosynthesis.
6. Separation of amino acids / pigments by paper chromatography.
7. To determine the absorption of water by Oily and starchy seed.

Distribution of marks (Practical):

Experiments	:	10 marks
Spotting	:	05 marks
Class records/Charts/Model/Herbarium etc.	:	05 marks
Viva voce	:	05 marks

SEMESTER-VIII

MJ-BOT-20 (DISSERTATION/SEMINAR)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **End Semester University Examination: 100 Marks**

Course Objectives:

1. To enhance students' communication skills through active participation in seminars and group discussions.
2. To prepare students for professional and academic settings where effective communication and collaboration are essential.

Course Outcomes:

1. Students will be prepared to apply the communication and collaboration skills developed in this course to professional and academic contexts, enhancing their overall employability and academic success.

Course Contents:

The Head of the Department will divide all the students of the class into different groups and each group will be allotted topics from different fields of Botany. Students will have to work under the supervision of a teacher of the department. Each and every student has to submit electronically typed hardbound dissertation along with the raw data, a week before the examination.

Steps of Evaluation:

1. Presentation and the report submitted by the students will be evaluated by one external member and one internal member.

External Member will be appointed by the Head of Department from the following:

1. Permanent professors working in the postgraduate department of the university or other colleges.
- OR
2. Retired Professor/Associate Professor/Assistant Professor of the university.

Distribution of Marks:

- | | |
|-----------------------|----------|
| • Dissertation Report | 50 Marks |
| • Presentation | 25 Marks |
| • Viva- voce | 25 Marks |

AMJ-BOT-01 (Advance Microbiology)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives:

1. To learn about the diversity of microbial world, kingdom and domain concept; features of dark field-, phase contrast.

Course Outcome:

1. The students will gain a holistic concept on history, development, scope and aspects of Microbiology.
2. They will also learn about the contributions of Microbiologists.
3. Students will be made aware of biosafety protocols and laboratory management.

Unit 1: History and Development of Microbiology: History, Development and Scope of microbiology; Contributions of Antonie van Leeuwenhoek, Louis Pasteur & Alexander Fleming in the field of Microbiology. **(10 Lectures)**

Unit 2: Basic Microscopy & Serology: Principle of Bright Field Microscope, Dark Field Microscope, Phase Contrast Microscope. **(10 Lectures)**

Unit 3: Microbial Taxonomy: Classification of microorganisms; Hackel's three kingdom concept – Whittaker's five kingdom concept – three domain concept of Carl Woese. **(10 Lectures)**

Unit 4: Viruses: Morphology, Ultrastructure & Chemistry of capsid and viral genome. **(05 Lectures)**

Unit 5: Elementary idea of virioids & prions. **(05 Lectures)**

Unit 6: Basic concept of immunology; interferon, microbial food infestation and food poisoning. **(10 Lectures)**

Unit 7: Role of microbes in agriculture, dairy & beverage industry. **(10 Lectures)**

Suggested readings:

1. Atlas RM. (1997). Principles of Microbiology. 2nd edition M. T. Brown Publishers.
2. Black JG. (2008). Microbiology: Principles and Explorations. 7th edition. Prentice Hall
3. Madigan MT, and Martinko JM. (2014). Brock Biology of Micro-organisms. 14th edition. Parker J. Prentice Hall International, Inc.
4. Pelczar Jr MJ, Chan ECS, and Krieg NR. (2004). Microbiology. 5th edition Tata McGraw Hill.
5. Srivastava S and Srivastava PS. (2003). Understanding Bacteria. Kluwer Academic Publishers, Dordrecht.

AMJ-BOT-02 (Advance Phytopathology)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course objectives: On completion of this course, the students will be able to understand

1. To introduce students with history and growth of plant pathology.
2. To gain knowledge of contribution of various pathologist in the field of plant pathology.
3. To gain knowledge of diseases development and diseases of plants and their control.

Course Outcome:

1. On successful completion of this course the student should know the clear views of general symptoms of different plant diseases.
2. The students will gain a holistic concept on history, development, scope and aspects of plant pathology.
3. They will also learn about the contributions of plant pathologist.

Unit 1: PLANT PATHOLOGY

1. **History & growth of plant pathology in modern period. (5 Lectures)**
2. **Contribution of various pathologists in the field of plant pathology. (5 Lectures)**
3. **Disease development:** Pathogenesis and Host-Parasite relationship. **(5 Lecture)**
4. **Role of enzymes in plant diseases. (5 Lectures)**
5. **Role of toxin in plant diseases. (5 Lectures)**
6. **Defense mechanisms:** Structural and Biochemical defense. **(5 Lecture)**
7. **Terms, classification, symptoms and transmission related to plant pathology. (5 Lectures)**
8. **Plant disease management and method:** Chemical, Biological and Cultural methods. Management methods: quarantine, disease clinics and disease forecasting (preliminary ideas). **(10 Lecture)**
9. **Plant Diseases:** A general account of Little leaf of Brinjal, Yellow vein mosaic of Bhindi, TMV, Citrus Canker, Late blight of potato, Black rust of wheat & Red rot of sugarcane. **(15 Lecture)**

Suggested Readings:

1. Mehrotra RS. 2006. Plant pathology. Tata McGraw Hill Publishing Co. Ltd., New Delhi.
2. Sharma PD. 2004. Plant pathology. Rastogi Publications, New Delhi.
3. Bilgrami S and Dubey HC. 1998. A text book of modern Plant pathology. Sangam Books Ltd., Mumbai.
4. Singh RS. 1990. Plant diseases. Oxford and IBH Publishing Co., New Delhi.
5. Butler EJ. 1973. Fungi and diseases in plants. Periodical Expert Book Agency, Delhi.
6. Rangaswamy G. 1972. Disease of crop plants in India. Prentice Hall of India, New Delhi.
7. Smith KM. 1968. Plant viruses. Methuen, London.
8. Mundkar BB. 1967. Fungi and Plant diseases. McMillan and Co. Ltd., Calcutta.

AMJ-BOT-03 (Practical based on AMJ-BOT-01 & 02)

- **Total Credits: -04**
- **Total teaching Hours: 120**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Unit 1: Microbiology Laboratory Management and Biosafety

Unit 2: To study the principle and applications of important instruments (autoclave, incubator, hot air oven, centrifuge, light microscope, pH meter) used in the microbiology laboratory.

Unit 3: Preparation of culture media and staining and identification of bacteria.

Unit 4: Sterilization of medium using Autoclave and assessment for sterility.

Unit 5: Sterilization of glassware using Hot Air Oven.

Unit 7: Preparation of pure culture of pathogens.

Unit 8: Spore measurement by stage and ocular micrometer.

Unit 9: Histo-pathological study of common diseases of local area.

Distribution of marks (Practical):

Experiments	:	30 marks
Spotting	:	10 marks
Class records/Charts/Model/Herbarium etc.	:	20 marks
Viva voce	:	15 marks

SEMESTER-VIII

GUIDELINE AND SYLLABUS FOR 4-YEAR UG DEGREE (HONOURS WITH RESEARCH)

Eligibility for Students:

Students who secure 75% marks and above in the first six semesters and wish to undertake research at the undergraduate level can choose a research stream in the fourth year. They should do a research project or dissertation under the guidance of a faculty member of the University/College. The research project/dissertation will be in the major discipline. The students, who secure 160 credits, including 12 credits from a research project/dissertation, are awarded UG Degree (Honours with Research).

Infrastructure Requirement for offering this Course:

The Departments offering a 4-year UG Degree (Honours with Research) must have the required infrastructure such as the laboratory, equipment's, glass wares, chemicals, class work materials, library, access to journals and computer lab facilities to carry out the research work, and at least two permanent faculty members who are recognized as Ph.D. supervisors. The Departments already recognized for conducting the Ph.D. programme may conduct a 4- year UG Degree (Honours with Research) without obtaining any approval from the affiliating University.

Semester	Course Category	Code	Papers	Credits
Eighth Semester	Research courses	RC-01	Research Methodology	4
	Research Project	RC-02	Research Project or Dissertation	8
			Total Credits	12

RC-BOT-01 (RESEARCH METHODOLOGY)

- **Total Credits: -04**
- **Total teaching Hours: 60**
- **Total Marks: 100**
- **Semester Internal Examination: 25 Marks**
- **End Semester University Examination: 75 Marks**

Course Objective:

- The course aims to inculcate research aptitude among the learners and equip them with knowledge and skills required to successfully undertake various steps in the research process.

Course Outcomes: After completion of the course, learners will be able to:

- Analyze research concepts, its types and steps in the research process.
- Formulate research problems and assimilate various types of research design.
- Create scales, sampling techniques and design data collection tools.
- Recognize various techniques of data analysis and interpretation.
- Prepare a complete research report in appropriate format.

Unit-1: Introduction of Research

Research: Meaning, Definition, Nature, Scope, Significance, Steps Types and Methods of Research. Characteristics of Goods Research **(15 Lectures)**

Unit-II: Research Process

Defining Research Problem, Title Formulation; Setting of Hypothesis, Research Design - Exploratory, Descriptive and Experimental Research Designs. **(15 Lectures)**

Unit-IV: Data

Data, types of data, Methods of Collection of Primary and Secondary Data. Process of Questionnaire Design; Processing of Data-Editing, Coding, Classification and Tabulation. **(10 Lectures)**

Unit-III: Sample and Sampling

Meaning of Sample, Sampling Design, Criteria of Selecting a Sampling Procedure, Characteristics of Good Sample Design. **(10 Lectures)**

Unit-V: Report Writing

Meaning and Types of Report, Steps of Report Writing, and Characteristics of a good report, Reference and bibliography. **(10 Lectures)**

Suggested Readings:

1. Research Methodology – Text and Cases with SPSS Applications, by Dr S.L. Gupta and Hitesh Gupta, International Book House Pvt Ltd
2. Methodology of Research in Sciences, by O.R. Krishnaswami, Himalaya Publishing House
3. Research Methodology, Methods and Techniques by C.R Kothari
4. Research Methodology by Dr Vijay Upagude and Dr Arvind Shende
5. Gupta, Santhosh Research Methodology and Statistical Techniques, Deep and Deep.
6. Dawson, C. (2002). Practical research methods. UBS Publishers, New Delhi.
7. Stapleton, P., Yondeowei, A., Mukanyange, J., Houten, H. (1995). Scientific writing for agricultural research scientists – a training reference manual. West Africa Rice Development Association, Hong Kong.
8. Ruzin, S.E. (1999). Plant micro technique and microscopy. Oxford University Press, New York, U.S.A

RC-BOT-02 (RESEARCH PROJECT OR DISSERTATION)

- **Total Credits: -08**
- **Total Marks: 200**

Objectives: The objectives of the course is to facilitate students to carry out extensive research and develop as self-guided learning and analytical skills through problem and gap identification, development of research methodology, interpretation of findings and presentation of results.

Course Outcomes: After completion of the course, the learners will be able to:

- Gain in-depth knowledge in the major field of study.
- Design and justify research methodology.
- Utilize appropriate research methodology for research.
- Analyze the data and draws conclusions accordingly.

Course Contents:

Students who secure 75% marks and above in the first six semesters and wish to undertake research at the undergraduate level can choose a research stream in the fourth year.

The HOD of the department has to allot supervisor to the students from among the permanent faculty members who have Ph.D. degree, after that the students have to select a research problem with the help of the supervisor and they have to submit a summary or research proposal to the department. Thereafter, the HOD of the department will organize a meeting of the Departmental Research Committee and after the presentation of the student the committee will approve or reject his/her synopsis/research proposal. Students will start their research work after getting approval from the department research committee.

At the end of the semester the student has to submit the project thesis to the department and after that the department will send that thesis to the external member, after receiving the report from the external member the department will conduct an open viva voce examination for the students in the presence of the external member as well as the internal member.

The Department Research Committee will be constituted from the following persons:

1. HOD of the Department–Chairmen
2. Dean, Faculty of Commerce- External member
3. HOD, University Department- External member
4. At list two permanent faculty member of the department (Nominated by HOD)- Members

External members can be any of the following:

1. Permanent professors working in the postgraduate department of the university or other colleges who have the qualification to become PhD supervisors.

OR

2. Retired Professor/Associate Professor/Assistant Professor of the university who has been supervise PhD scholar.

OR

3. Professor/Associate Professor/Assistant Professor of the outside university who has been supervise PhD scholar.

Note- Minimum three external examiner lists will be sent by HOD through concern college principal to controller of examination, SKMU, Dumka for final approval. In that list priority will be given as per order mentioned above.

The project thesis will evaluate under the following heads:

- Motivation for the choice of topic
- Project dissertation design
- Methodology and Content depth
- Results and Discussion
- Future Scope & References
- Participation in Internship programme with reputed organization
- Application of Research technique in Data collection
- Report Presentation
- Presentation style Broad

Guidelines for distribution of marks may be as follows or as appropriate:

- Assessment of project synopsis: 75 marks
- Assessment of project thesis: 100 marks
- Viva-voce: 25 marks

A collection of handwritten signatures and dates, likely representing the approval of external examiners. The signatures are written in various colors (black, blue, red) and include the following names and dates:

- Top Left:** A signature with a checkmark and the date 27/04/24.
- Top Middle:** A signature in red ink with the date 27.04.24.
- Top Right:** A signature in black ink with the date 27/4/24.
- Middle Left:** A signature in black ink with the date 27/04/24.
- Middle Middle:** A signature in blue ink with the date 27.4.24.
- Middle Right:** A signature in black ink with the date 27/4/24.
- Bottom Left:** A signature in black ink with the date 27/04/2024.
- Bottom Middle:** A signature in black ink with the date 27/04/2024.
- Bottom Right:** A signature in black ink with the date 27/04/24.
- Bottom Far Right:** A signature in black ink with the date 27/04/24.
- Bottom Left (Bottom Row):** A signature in black ink with the date 27/4/2024.